

WATER SUPPLY SYSTEM FOR WASH BASIN

BASIC WATER SUPPLY AND DRAINAGE ARRANGEMENT

PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree

BACHELOR OF ENGINEERING

in

CHEMICAL ENGINEERING

SAVEETHA ENGINEERING COLLEGE,

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SEPTEMBER 2026

ABSTRACT

Water supply and plumbing systems are essential components of residential, commercial and institutional buildings. A properly designed water supply system ensures the safe and efficient delivery of water from a storage tank to fixtures such as wash basins. This project presents a simple water supply arrangement for a wash basin. The system consists of a water tank connected to a pipeline through a gate valve. Elbows are used to change the direction of the pipe and another valve is provided to control the supply before the tap. Water is delivered through the tap into the wash basin, while wastewater is discharged through a flexible drain pipe.

Two drawing methods are included: the double line diagram and the single line diagram. The double line diagram gives a more detailed representation of the physical pipe arrangement, while the single line diagram provides a simplified schematic representation of the same system. The report explains the function of each component, water flow sequence, installation procedure, advantages, maintenance and safety considerations.

Keywords: Water supply, plumbing, water tank, gate valve, elbow, tap, wash basin, drain pipe, single line diagram, double line diagram.

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1. INTRODUCTION, AIM AND OBJECTIVES

1.1 Introduction

Water is an essential requirement in every building. A building water supply system is designed to convey water from a source or storage tank to different points of consumption. Plumbing systems are therefore an important part of building services and must be properly planned, installed and maintained.

The given project represents a basic water supply connection from a water tank to a wash basin. Water stored in the tank flows through a supply pipe. A gate valve is provided near the tank to control or isolate the water supply. The pipeline then changes direction through elbows and continues towards the wash basin. Another valve is provided along the pipeline before the tap. When the tap is opened, water flows into the wash basin. Wastewater generated during use is collected through the basin outlet and discharged using a flexible drain pipe.

1.2 Aim

To study and represent a basic water supply and drainage arrangement for a wash basin, identify its major components and describe a suitable installation and maintenance procedure.

1.3 Objectives

- Understand the basic arrangement of a building water supply system.
- Represent the connection between a water tank and wash basin.
- Understand the function of a gate valve and control valve.
- Understand the purpose of elbows in changing pipeline direction.
- Study the function of a tap as the point-of-use water outlet.
- Understand the function of a wash basin in collecting used water.
- Provide a suitable drainage arrangement for wastewater.
- Understand the difference between single line and double line diagrams.
- Understand the sequence of installation of pipes and plumbing fittings.
- Identify the major components required for the plumbing arrangement.
- Understand the importance of controlling water flow using valves.
- Study basic plumbing maintenance and safety requirements.

1.4 Scope

This project covers the basic water supply path, drainage arrangement, component functions, schematic representation, installation sequence, maintenance and safety considerations for a wash basin connection.

2. COMPONENTS AND ARRANGEMENT

2.1 Major components

S. No. Component Function / Description

- 1 Water Tank Stores water and supplies it to the plumbing line.
- 2 Gate Valve Starts, stops, or isolates the water supply.
- 3 Water Supply Pipe Carries water from the tank to the wash basin.
- 4 Elbow Changes the direction of the pipeline.
- 5 Control Valve Controls or isolates the water supply before the tap.
- 6 Tap Delivers water into the wash basin and controls the flow.
- 7 Wash Basin Receives water from the tap and collects used water.
- 8 Flexible Drain Pipe Carries wastewater from the basin to the drainage system.

2.2 Water flow arrangement

The overall flow sequence is: Water Tank → Gate Valve → Elbows / Supply Pipe → Control Valve → Tap → Wash Basin → Flexible Drain Pipe.

If the tank is elevated above the fixture, gravity can assist the movement of water through the supply pipe. The gate valve provides line isolation, while the control valve provides local control before the tap.

2.3 Arrangement concept

The supply side and drainage side form one simple building-service arrangement. The supply pipe conveys clean water to the fixture, and the basin outlet with flexible drain pipe carries used water away.

3. PROPOSED DRAWING VIEWS

The project uses two drawing methods to communicate the same plumbing arrangement. The double line drawing emphasizes the physical pipe arrangement, while the single line drawing emphasizes the flow path.

3.1 Double line representation

A double line diagram represents the plumbing pipe using two lines and gives a more realistic indication of the physical installation. It makes the location of valves, elbows and fixtures easier to visualize.

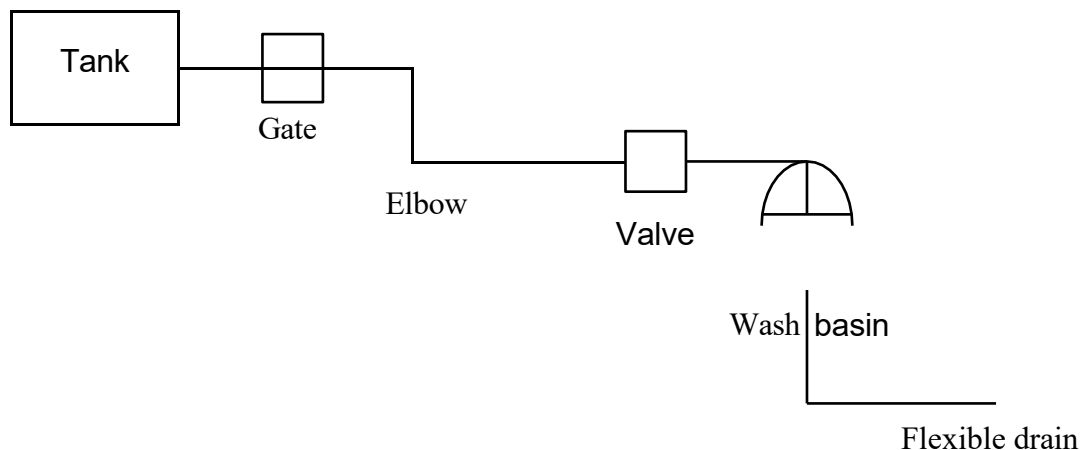
3.2 Single line representation

A single line diagram is a simplified schematic representation. One line represents the pipe, making the drawing easy to read and suitable for schematic layouts and preliminary planning.

3.3 Drawing interpretation

View / representation	Main purpose
Double line diagram	Shows the physical pipe arrangement, fittings and fixture positions more
Single line diagram	Shows the simplified flow path and overall plumbing relationship.
Front / elevation view	Can show the relative vertical positions of tank, pipe route and wash basin
Sectional view	Can show pipe, basin outlet and drainage connection where required

Single-line schematic of water supply and drainage



4. WORKING AND INSTALLATION PROCEDURE

4.1 Proposed installation procedure

Step 1 – Water Storage – Water is stored inside the water tank. The tank should be properly supported and protected from contamination.

Step 2 – Gate Valve Installation – A gate valve is installed at the tank outlet so that the complete supply line can be isolated when required.

Step 3 – Pipe Connection – The supply pipe is connected to the gate valve. Pipe joints should be properly sealed to prevent leakage.

Step 4 – Direction Change – Elbows are installed wherever the pipe route changes direction.

Step 5 – Control Valve Installation – A second valve is provided before the tap to control or isolate the fixture supply.

Step 6 – Tap Connection – The pipe is connected to the tap, which is fixed securely above the wash basin.

Step 7 – Wash Basin Installation – The basin is fixed at a suitable height and its outlet is connected to the drainage arrangement.

Step 8 – Drain Pipe Connection – A flexible drain pipe is connected below the basin outlet to carry wastewater away.

Step 9 – Testing – The supply is opened gradually and all joints are checked for leakage.

Step 10 – Final Operation – The tap is opened to check water flow, and the drain is checked to confirm free discharge.

4.2 Working principle

Water stored in the tank enters the supply pipe when the gate valve is open. The pipeline follows the required route using elbows. The control valve permits local isolation or control before the tap. When the tap is opened, water is delivered into the wash basin. Used water leaves through the basin outlet and flexible drain pipe.

5. SCHEMATIC DIAGRAMS

Figure 1. Double line representation of the water supply and drainage arrangement.

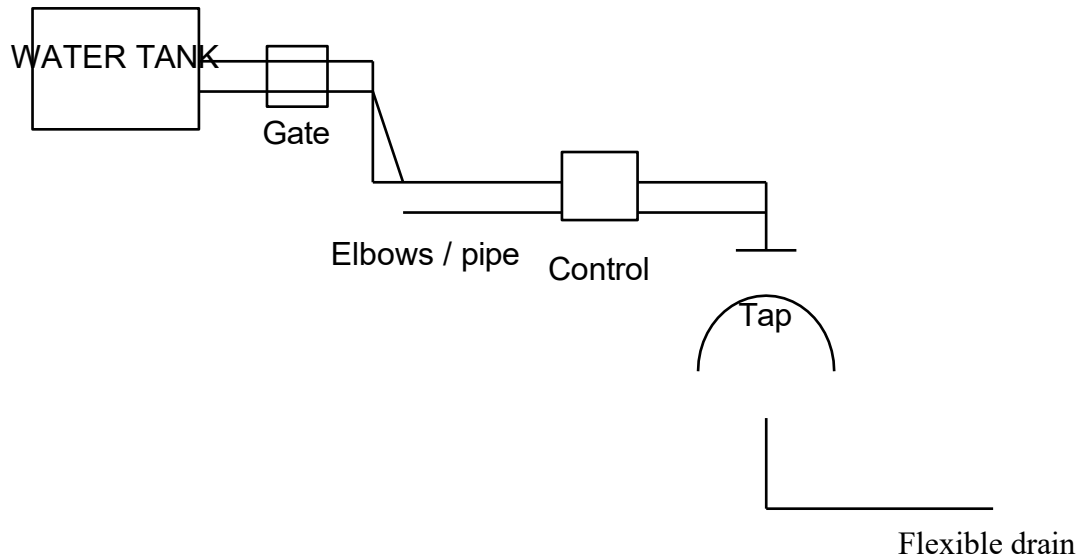
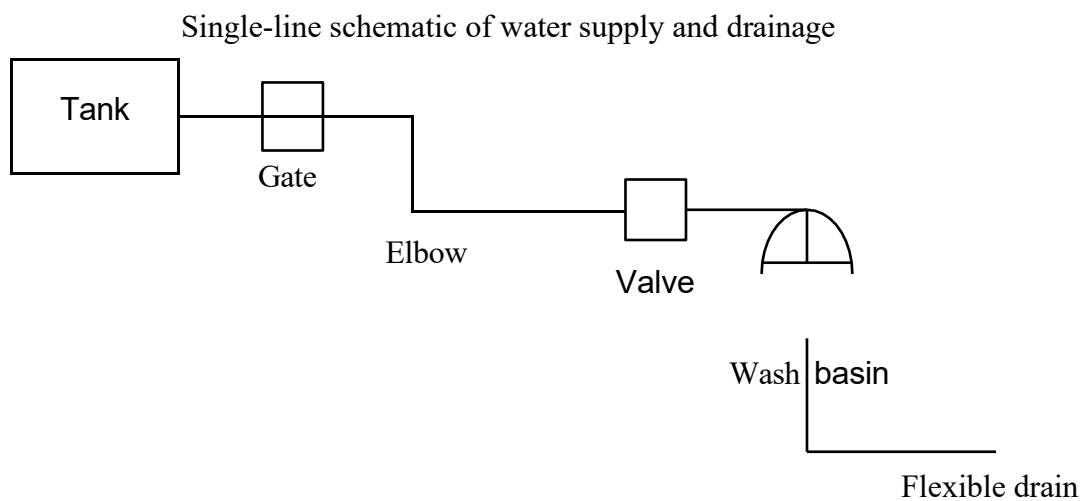


Figure 2. Single line schematic of the water supply and drainage



The diagrams are schematic representations prepared from the project description. The double line view emphasizes the physical pipe arrangement; the single line view emphasizes the flow path.

6. EVALUATION, MAINTENANCE AND SAFETY

6.1 Evaluation

Item	Check / observation
Pipe joints	Check periodically for leakage and confirm that connections remain sealed.
Valves	Operate periodically to ensure proper functioning and convenient isolation.
Tap and basin	Clean the tap outlet and basin regularly.
Drainage	Check the basin outlet and flexible drain pipe for blockage or leakage.
Water tank	Inspect for cleanliness and protect stored water from contamination.

6.2 Advantages

- Simple and easy-to-understand plumbing arrangement.
- Gate valves allow convenient isolation of the supply line.
- Elbows provide flexibility in routing the pipe.
- The tap gives convenient control at the point of use.
- The flexible drain pipe provides a simple connection to the drainage system.
- The single line diagram is easy to read and suitable for schematic planning.
- The double line diagram provides a clearer physical representation.

6.3 Safety precautions

- Ensure that the water tank is securely supported.
- Keep pipe connections properly sealed.
- Repair leakage immediately.
- Keep electrical equipment away from leaking water.
- Protect stored water from contamination.
- Operate valves and taps without excessive force.

7. CONCLUSION AND REFERENCE MATERIAL

7.1 Conclusion

The project presents a simple and practical water supply and drainage arrangement for a wash basin. The system consists of a water tank, gate valve, supply pipe, elbows, control valve, tap, wash basin and flexible drain pipe. The water tank acts as the storage source and the pipeline conveys water towards the wash basin. The gate valve and control valve provide control and isolation of the water supply. Elbows allow the pipeline to follow the required route. The tap delivers water to the wash basin, while the flexible drain pipe carries wastewater away from the basin.

The double line diagram provides a detailed representation of the physical plumbing arrangement, whereas the single line diagram provides a simplified schematic representation. Both methods are useful for plumbing design and communication. Proper installation, testing and maintenance are essential for achieving a reliable and leak-free plumbing system.

7.2 Reference material

1. Bureau of Indian Standards (BIS) — National Building Code of India 2016 (NBC 2016), Part 9: Plumbing Services, Section 1: Water Supply and Section 2: Drainage and Sanitation — 2016.
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